

X is a random variable describing some physical characteristic.
 Assume there is some underlying probability model for X , i.e.,
 X has a probability distribution with pmf/pdf f_θ .
 θ is a parameter (unknown scalar or vector), which characterises the
 distribution of X . θ belongs to some parameter space Θ , say.

e.g., $X \sim \text{Bin}(n, \theta)$, $\theta \in (0, 1)$, $\Theta = \{\theta: \theta \in (0, 1)\}$
 $X \sim P(\theta)$, $\theta > 0$ ← similarly for others
 $X \sim \text{Exp}(\theta)$, $\theta > 0$
 $X \sim N(\mu, \sigma^2)$, $\Theta = \{(\mu, \sigma^2): \mu \in \mathbb{R}, \sigma^2 > 0\}$

To estimate or test hypotheses
 The goal of statistical inference (parametric) about unknown parameter/s.
 We collect, store, conserve large amount of data for this purpose.
 $x_1, \dots, x_n$ is a random sample from f_θ . We use this sample to
 infer about θ .

Can we summarize the data on $x_1, \dots, x_n$ w/o losing information about
 the parameter? The summarizing statistic is a sufficient statistic.
 The whole idea is 'data dimension reduction w/o loss of information'.

As the name suggests, a statistic $T(x_1, \dots, x_n)$ based on the r.s. $x_1, \dots, x_n$
 is sufficient for θ (or, f_θ) if it contains all info' on θ , that is
 contained in the entire sample. Once a sufficient statistic is known, other
 details about the data tell us nothing extra about the unknown parameter.

e.g., $x_1, \dots, x_n$ iid $\text{Ber}(\theta)$
 Full data contains n 0's and 1's
 Instead, it is enough to know $\sum_{i=1}^n x_i$ (# of zeros or ones, depending on the def'n of x_i)
 (why?)

Definition Let the random sample $x_1, \dots, x_n$ has a jth distribution dependent
 on θ . Then $T(x_1, \dots, x_n)$ is sufficient for θ , if the conditional
 distribution of the data given $T=t$ is independent of θ .
 $(x_1, \dots, x_n)$
 $[P_\theta(x_1=x_1, \dots, x_n=x_n | T=t)]$ discrete case, and $f_\theta(x_1, \dots, x_n | T=t)$ continuous case
 do not depend on θ .

Ex 1 $X_1, \dots, X_n$ r.s. from $\text{Ber}(\theta)$, $0 < \theta < 1$.

Claim $T = \sum_{i=1}^n X_i$ is sufficient for θ . Note $T \sim \text{Bin}(n, \theta)$.

$$P(X_1=x_1, \dots, X_n=x_n | T=t) = \frac{P(X_1=x_1, \dots, X_n=x_n, T=t)}{P(T=t)} = \begin{cases} \frac{P(X_1=x_1, \dots, X_n=x_n)}{P(T=t)} & \text{if } \sum x_i = t \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{\theta^{x_1}(1-\theta)^{1-x_1} \dots \theta^{x_n}(1-\theta)^{1-x_n}}{\binom{n}{t} \theta^t (1-\theta)^{n-t}} & \text{if } \sum x_i = t \\ 0 & \text{otherwise} \end{cases}$$

$$= \begin{cases} \frac{1}{\binom{n}{t}} & \text{if } \sum x_i = t \\ 0 & \text{otherwise} \end{cases} \quad \text{does not depend on } \theta.$$

$\Rightarrow T = \sum_{i=1}^n X_i$ is sufficient for θ .

Ex 2 $X_1, \dots, X_n$ r.s. from $P(\lambda)$, $\lambda > 0$

$T = \sum_{i=1}^n X_i$ suff. for λ , $T \sim P(n\lambda)$

$$P(X_1=x_1, \dots, X_n=x_n | T=t) = \frac{e^{-n\lambda} \frac{(n\lambda)^{x_1+\dots+x_n}}{x_1! \dots x_n!}}{e^{-n\lambda} \frac{(n\lambda)^t}{t!}} \quad (\text{follow the steps above})$$

$$= \begin{cases} \frac{t!}{n^t \prod_{i=1}^n x_i!} & \text{if } \sum x_i = t \\ 0 & \text{otherwise} \end{cases}$$

does not depend on λ .

$\Rightarrow T = \sum_{i=1}^n X_i$ is sufficient for λ .

Note: The definition is ~~not~~ constructive. Works only for checking whether a statistic is sufficient or not. May not be easy to guess a sufficient statistic in many complex distributions.

Following is a practical tool to obtain sufficient statistics.

The Neyman-Fisher Factorization Theorem (NFFT)

$X_1, \dots, X_n$ is a random sample with pmf/pdf f_θ , $\theta \in \Theta$.

A statistic $T(X)$ is sufficient for θ iff $f_\theta(z)$ can be factorized as

$$f_\theta(z) = h(z) g_\theta(t(z)), \text{ where}$$

- (i) $h(z) > 0$ is a function of $(x_1, \dots, x_n)$ not depending on θ
- (ii) g_θ is a function depending on θ , and the data, but only through $t(z)$.

$g_\theta(t(z))$ is

Note

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- (i) The factorization is not unique
- (ii) So sufficient statistic is also not unique (checking through examples).
- (iii) Every function of a sufficient statistic is not ~~not~~ sufficient (Why??)
- (iv) Every one-to-one fn of a sufficient statistic is sufficient

eg-3 Finding suff. stat. by using NFFT

$x_1, \dots, x_n$ iid $N(\theta, 1)$, $\theta \in \mathbb{R}$.

the jlt pdf $f_{\theta}(x) = \frac{1}{(\sqrt{2\pi})^n} \exp\left\{-\frac{1}{2} \sum_{i=1}^n (x_i - \theta)^2\right\}$, $-\infty < x_1, \dots, x_n < \infty$.

$$= \frac{1}{(\sqrt{2\pi})^n} \exp\left(-\frac{1}{2} \sum x_i^2\right) \exp\left(\theta \sum x_i - \frac{n}{2} \theta^2\right)$$

$$= \underbrace{\frac{1}{(\sqrt{2\pi})^n} \exp\left(-\frac{1}{2} \sum x_i^2\right)}_{h(x)} \underbrace{\exp\left(\theta \sum x_i - \frac{n}{2} \theta^2\right)}_{g_{\theta}(t(x))} \text{ with } t(x) = \sum x_i.$$

However, $\therefore$ By NFFT, $T = \sum_{i=1}^n x_i$ is suff. for θ .

- (i) The factorization is possible in different ways, so that each of $\sum x_i$, $\bar{x}$, $(x_1, \sum_{i=2}^n x_i)$, $(\sum_{i=1}^{2m} x_i, \sum_{i=2m+1}^n x_i)$, $(x_1, \dots, x_n)$ is a sufficient statistic.
- (ii) We prefer the one which gives maximum reduction of data dimension, i.e. $\sum x_i$ (or $\bar{x}$). The entire data $(x_1, \dots, x_n)$, with no reduction in dim. is the trivial sufficient statistic.

eg-4 Check $x_1, \dots, x_n$ iid $P(\lambda)$ by the NFFT

eg-5 $x_1, \dots, x_n$ r.v. from $U(0, \theta)$, $\theta > 0$.

joint pdf $f_{\theta}(x_1, \dots, x_n) = \begin{cases} \frac{1}{\theta^n} & \text{if } 0 < x_1, \dots, x_n < \theta \\ 0 & \text{or} \end{cases}$

$$= \begin{cases} \frac{1}{\theta^n} & \text{if } 0 < x_{(1)} < x_{(n)} < \theta \\ 0 & \text{or} \end{cases}$$

$$= \frac{1}{\theta^n} I(0, x_{(n)}) I(x_{(n)}, \theta)$$

* NFFT with $h(x) = I(0, x_{(n)})$

$$g_{\theta}(t(x)) = \frac{1}{\theta^n} I(x_{(n)}, \theta)$$

Indicator variable

$$I(a, b) = \begin{cases} 1 & \text{if } a \leq b \\ 0 & \text{if } a > b \end{cases}$$

$\Rightarrow T(x) = x_{(n)}$ is sufficient for θ .